Chapter 7

Population means and significance

- Randomization/permutation test
  - If there’s no difference, we can shuffle the data together
  - Have to do each combination

t-statistic

- Difference divided by the standard error of the difference of means:
  - $t_s = \frac{\bar{y}_1 - \bar{y}_2}{SE_{(\bar{Y}_1 - \bar{Y}_2)}}$
- If the population means are the same, then the t-statistic has the Student's t distribution

Hypothesis testing: t-test

- Compare population means from 2 independent samples
- H_0: $\mu_1 = \mu_2 \Leftrightarrow \mu_1 - \mu_2 = 0$
  - They're the same
- H_A: $\mu_1 \neq \mu_2 \Leftrightarrow \mu_1 - \mu_2 \neq 0$
  - They're different

P-value

- If the null hypothesis is true, the p-value is the probability of observing a statistic as extreme or more extreme than was observed
- Exact probabilities of a statistic are low or 0
- Two-tailed tests are usually done because we don’t care about the order of the means (i.e. which one is larger or smaller)

P-value

- We pick a significance level, α, before seeing the data
  - Usually, 0.05 or 0.01
- This is the threshold we'll use for deciding what's "unlikely"
  - If the p-value is greater than α, then we cannot reject the null
  - If the p-value is less than α, then we reject the null
  - E.g. α = 0.05
    - With a p-value of 0.25, we cannot reject the null hypothesis (that they're equal)
    - With a p-value of 0.04, we reject the null hypothesis in favor of the alternative (not equal)
- This is now a question about the probability of the difference being big or not instead of the difference in sample means being big or not

- If we reject the null hypothesis, 2 possibilities:
  - Null is false: **correct** conclusion
  - Null is true: **wrong** conclusion
    - This is a false positive – we reject when we shouldn't have
    - We got a relatively rare dataset and the p-value quantifies that
      - Previous example: p = 0.04 or 4% chance of getting this outcome if the null is true

$\bar{y}_1 - \bar{y}_2$ or $\bar{y}_2 - \bar{y}_1$

- For non-directional alternative hypothesis it doesn’t matter
  - The signs will change for the confidence interval but same conclusion
  - When we compute the t-statistic we take the absolute value

- If we don’t reject the null hypothesis, 2 possibilities:
  - Null is true: **correct** conclusion
  - Null is false: **wrong** conclusion
    - This is a false negative – we didn’t reject when we should have
      - Can be due to small difference or small sample size

Statistical significance

- Depends on sample size
- Not always clear
- Best practice to report both p-value and confidence interval

Effect size

- $|\mu_1 - \mu_2|/\sigma$
- Can be approximated by:
  - $|\bar{y}_1 - \bar{y}_2|/\max(s_1, s_2)$

t-tests and confidence intervals

- Different ways to use the same information
- t-tests give p-values which are strength of evidence that $\mu_1 \neq \mu_2$
- Confidence intervals can show the magnitude of difference

Significance level α and p-value

- Both are tail areas in Student's t distribution
- Fix significance level α at the beginning
- Calculate p-value based on data

P-value

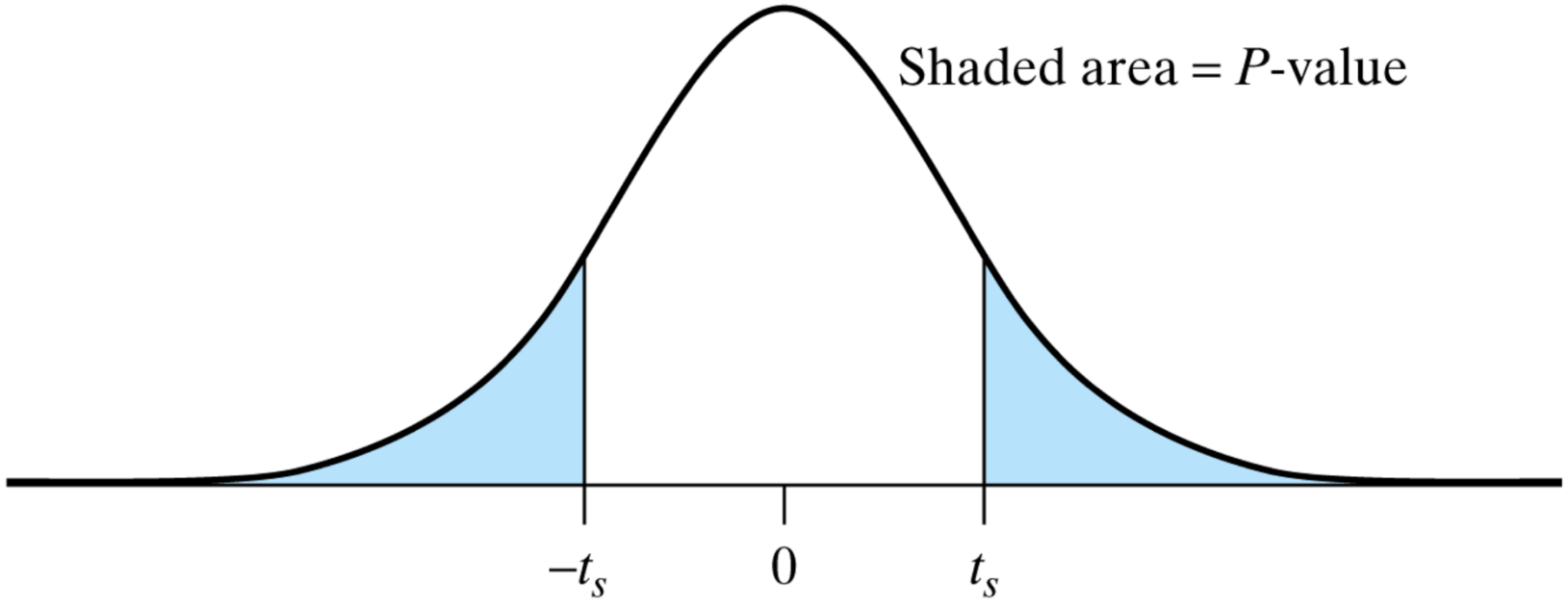
- Assuming the null hypothesis is true (the population means are equal), this is the probability of observing a result as extreme or more extreme than what was observed
- Null hypothesis is assumed to be true and then we calculate

P-value for the t-test

- We assume the population mean is the same for the t-test
- Then, the p-value is the probability that

$$|t_s| = |\bar{y}_1 - \bar{y}_2| / SE_{(\bar{Y}_1 - \bar{Y}_2)}$$

is as large or larger than observed



P-value

- It is not the probability the null hypothesis is true
- Type I error: False positive, reject H_0 when H_0 is true
- Significance level $\alpha = P(reject\ H_0 | H_0\ is\ true)$
- We protect against Type I errors by choosing α

Consequences of Type I and Type II errors

- For example,

  H_0: new treatment has no effect

  H_A: new treatment has an effect

- Type I error (false positive)

  We find significant evidence that a new treatment has an effect, when in fact the new treatment has no effect

- Type II error (false negative)

  We find no significant evidence that a new treatment has an effect, when in fact the new treatment does have an effect

Power

- Power of a statistical test = P(reject H_0 | H_A is true)
- Depends on
  - Magnitude of difference between $\mu_1 \; and \; \mu_2$
  - Sample sizes (increases with larger sample size)
  - Statistical test
- If the null is true, at $\alpha = 0.05$, there is still a 5% chance the p-value is less than 0.05

One-tailed t-test

- Sometimes we want the alternative hypothesis to be one population mean is larger than the other (instead of just being different)
- Can make the difference between rejecting or not rejecting
- Decide before seeing the data
- $(\bar{y}_B - \bar{y}_S) - t_\alpha SE_{\bar{Y}_B - \bar{Y}_S}$
  - If 0 is included in the confidence interval, we cannot reject the null
  - If 0 is not included in the confidence interval, we can reject the null

t-test and confidence intervals

- 90%
  - If the 90% confidence interval does not include 0, we can **reject the null** hypothesis at significance level $\alpha = 0.1$
  - If the 90% confidence interval does include 0, we **cannot reject the null** hypothesis at significance level $\alpha = 0.1$
- 95%
  - If the 95% confidence interval does not include 0, we can **reject the null** hypothesis at significance level $\alpha = 0.05$
  - If the 95% confidence interval does include 0, we **cannot reject the null** hypothesis at significance level $\alpha = 0.05$
- 99%
  - If the 99% confidence interval does not include 0, we can **reject the null** hypothesis at significance level $\alpha = 0.01$
  - If the 99% confidence interval does include 0, we **cannot reject the null** hypothesis at significance level $\alpha = 0.01$

Conditions for t-test

- The 2 samples must be simple random samples
- The two samples must be chosen independently of each other
- The distribution of means must be approximately normal
  - If sample sizes are large (>20), CLT starts guaranteeing this
  - Small sample sizes require the samples be normally distributed
  - Look for outliers regardless of sample sizes
    - QQ plots

What if the conditions aren’t met for t-test

- Needs to be normal
- Check if the data is normal after transforming (e.g. log)
  - QQ plots to verify
- You can check if the log of the data is normal and perform a t-test on the log of the data

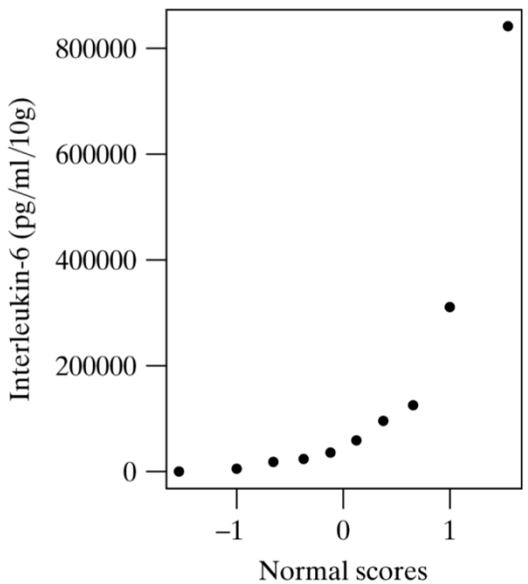
Example

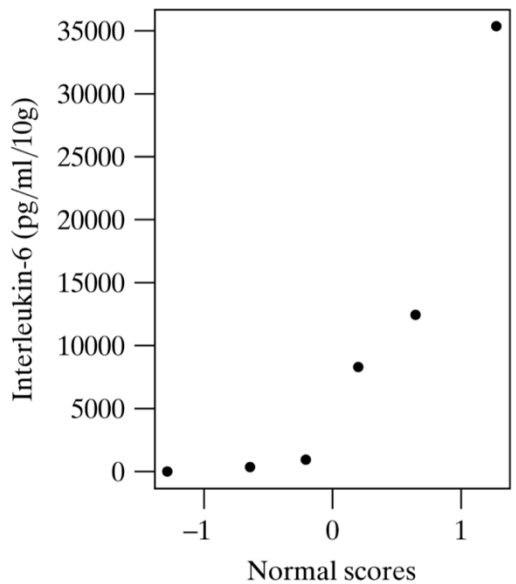
Small sample size, not normally distributed

Table 7.8.1 Interleukin-6 levels of breast implant patients and control patients

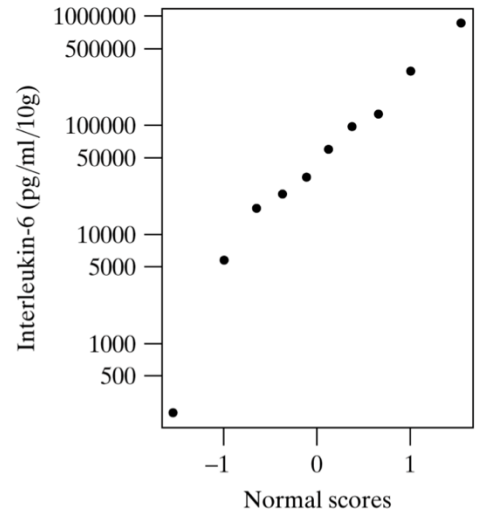
	Original data		Log scale	
	Breast implant patients	Control patients	Breast implant patients	Control patients
	231	35,324	2.364	4.548
	308,287	12,457	5.489	4.095
	33,291	8,276	4.522	3.918
	124,550	44	5.095	1.643
	17,075	278	4.232	2.444
	22,955	840	4.361	2.924
	95,102		4.978	
	5,649		3.752	
	840,585		5.925	
	58,924		4.770	
$\bar{y}$	150,665	9,537	4.549	3.262
s	259,189	13,613	0.992	1.111

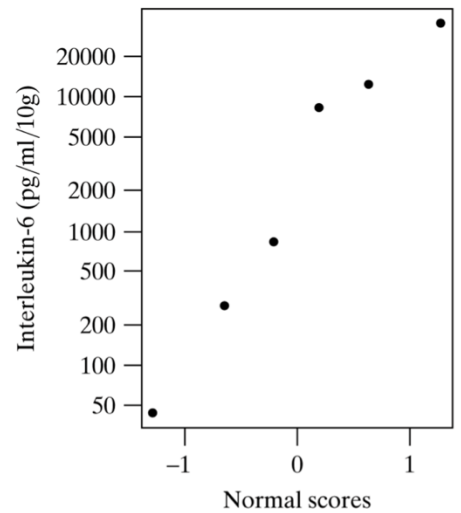
Normal quantile plot





Normal quantile plot of log(data)





What if the data and transform are not normal?

- Compare two independent, not-normal distributions
  - Null: the distributions are the same
  - Alternative: the distributions are different

Wilcoxon/rank sum test

- Not good when there are many ties
- Non-parametric, no assumptions on distribution or parameters
- Also known as
  - Wilcoxon-Mann-Whitney test
  - Mann-Whitney U test
  - Wilcoxon Test
  - Rank Sum Test

Example

- e.g., Soil respiration is a measure of microbial activity in soil (mol CO2/g soil/hr). Compare “Growth” location vs. “Gap” location

Growth location	Gap location
17, 20, 170, 315, 22, 190, 64	22, 29, 13, 16, 15, 18, 14, 6

Example

- Step 1: sorting data

Growth	# gap smaller	Gap	# growth smaller
17	5	6	0
20	6	13	0
22	6.5	14	0
64	8	15	0
170	8	16	0
190	8	18	1
315	8	22	2.5
		29	3

- Step 2
  - K_1 = sum(# gap smaller) = 49.5
  - K_2 = sum(# growth smaller) = 6.5
- Step 3
  - Checking: $K_1 + K_2 = 56$
  - $N_1 \times N_2 = 7 \times 8 = 56$
- Step 4
  - $U_s = \max(K_1, K_2)$
    $= \max(49.5, 6.5) = 49.5$

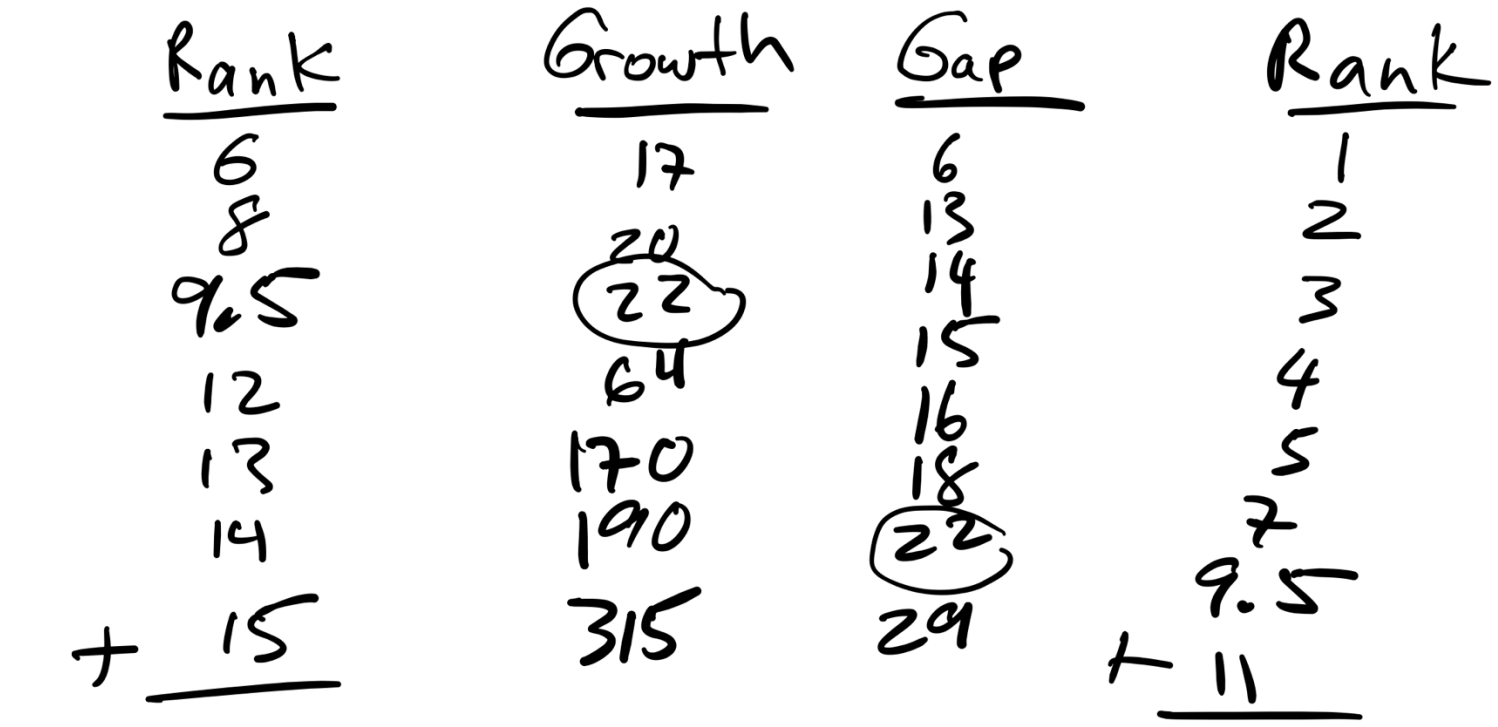
    (lookup table)
  - p between 0.0093 and 0.021
  - Reject null and accept alternative

TABLE 6 Critical values and *P*-values of U_s for the Wilcoxon-Mann-Whitney test

Note: Because the Wilcoxon-Mann-Whitney test null distribution is discrete, this table provides selected values of the test statistic U_s **in bold type** and corresponding *P*-values for a non-directional alternative *in italics*. Directional *P*-values are found by dividing the numbers in italics in half.

n	n'	0.20	0.10	0.05	0.025	0.01	0.005
3	2	**6** *0.200*					
	3	**8** *0.200*	**9** *0.100*				
4	2	**8** *0.133*					
	3	**11** *0.114*	**12** *0.057*				
	4	**13** *0.200*	**15** *0.057*	**16** *0.029*			
5	2	**9** *0.191*	**10** *0.095*				
	3	**13** *0.143*	**14** *0.071*	**15** *0.036*			
	4	**16** *0.191*	**18** *0.064*	**19** *0.032*	**20** *0.016*		
	5	**20** *0.151*	**21** *0.095*	**23** *0.032*	**24** *0.016*	**25** *0.0079*	
6	2	**11** *0.143*	**12** *0.071*				
	3	**15** *0.167*	**16** *0.095*	**17** *0.048*	**18** *0.024*		
	4	**19** *0.171*	**21** *0.067*	**22** *0.038*	**23** *0.019*	**24** *0.0095*	
	5	**23** *0.178*	**25** *0.082*	**27** *0.030*	**28** *0.017*	**29** *0.0087*	**30** *0.0043*
	6	**27** *0.180*	**29** *0.093*	**31** *0.041*	**33** *0.015*	**34** *0.0087*	**35** *0.0043*
7	2	**13** *0.111*	**14** *0.056*				
	3	**17** *0.183*	**19** *0.067*	**20** *0.033*	**21** *0.017*		
	4	**22** *0.164*	**24** *0.072*	**25** *0.042*	**26** *0.024*	**28** *0.0061*	
	5	**27** *0.149*	**29** *0.073*	**30** *0.048*	**32** *0.018*	**34** *0.0051*	**35** *0.0025*
	6	**31** *0.181*	**34** *0.073*	**36** *0.035*	**37** *0.022*	**39** *0.0082*	**40** *0.0047*
	7	**36** *0.165*	**38** *0.097*	**41** *0.038*	**43** *0.018*	**45** *0.0070*	**46** *0.0041*
8	2	**14** *0.178*	**15** *0.089*	**16** *0.044*			
	3	**19** *0.194*	**21** *0.085*	**22** *0.049*	**23** *0.024*		
	4	**25** *0.154*	**27** *0.073*	**28** *0.049*	**30** *0.016*	**31** *0.0081*	**32** *0.0040*
	5	**30** *0.171*	**32** *0.093*	**34** *0.045*	**36** *0.019*	**38** *0.0062*	**39** *0.0031*
	6	**35** *0.181*	**38** *0.081*	**40** *0.043*	[illegible]	[illegible]	**45** *0.0047*
	7	**40** *0.189*	**43** *0.094*	**46** *0.041*	**48** *0.021*	**50** *0.0093*	**52** *0.0037*
	8	**45** *0.195*	**49** *0.083*	**51** *0.050*	**54** *0.021*	[illegible]	**58** *0.0047*
9	2	**16** *0.146*	**17** *0.073*	**18** *0.036*			
	3	**22** *0.146*	**23** *0.100*	**25** *0.036*	**26** *0.018*	**27** *0.0091*	
	4	**27** *0.199*	**30** *0.076*	**32** *0.034*	**33** *0.020*	**35** *0.0056*	**36** *0.0028*
	5	**33** *0.190*	**36** *0.083*	**38** *0.042*	**40** *0.019*	**42** *0.0070*	**43** *0.0040*
	6	**39** *0.181*	**42** *0.088*	**44** *0.050*	**47** *0.018*	**49** *0.0076*	**50** *0.0048*
	7	**45** *0.174*	**48** *0.091*	**51** *0.042*	**53** *0.023*	**56** *0.0079*	**58** *0.0033*
	8	**50** *0.200*	**54** *0.093*	**57** *0.046*	**60** *0.021*	**63** *0.0079*	**65** *0.0037*
	9	**56** *0.190*	**60** *0.094*	**64** *0.040*	**66** *0.024*	**70** *0.0078*	**72** *0.0040*
10	2	**17** *0.182*	**19** *0.061*	**20** *0.030*			
	3	**24** *0.161*	**26** *0.077*	**27** *0.049*	**29** *0.014*	**30** *0.0070*	
	4	**30** *0.188*	**33** *0.076*	**35** *0.036*	**36** *0.024*	**38** *0.0080*	**39** *0.0040*
	5	**37** *0.165*	**39** *0.099*	**42** *0.040*	**44** *0.019*	**46** *0.0080*	**47** *0.0047*
	6	**43** *0.181*	**46** *0.093*	**49** *0.042*	**51** *0.023*	**54** *0.0075*	**55** *0.0047*
	7	**49** *0.193*	**53** *0.088*	**56** *0.043*	**58** *0.025*	**61** *0.0097*	**63** *0.0046*
	8	**56** *0.173*	**60** *0.083*	**63** *0.043*	**66** *0.021*	**69** *0.0085*	**71** *0.0044*
	9	**62** *0.182*	**66** *0.095*	**70** *0.044*	**73** *0.022*	**77** *0.0076*	**79** *0.0041*
	10	**68** *0.190*	**73** *0.089*	**77** *0.043*	**80** *0.023*	**84** *0.0089*	**87** *0.0039*

Why is it called rank sum?



$U_1 = R_1 - \frac{n_1(n_1+1)}{2}$

$= 77.5 - \frac{7(8)}{2}$

$= \boxed{49.5}$

$U_2 = R_2 - \frac{n_2(n_2+1)}{2}$

$= 42.5 - \frac{8(9)}{2}$

$= \boxed{6.5}$

Always get the same two values as the other method!

Example of directional alternative hypothesis

- E.g. H_A:Soil respiration is greater in growth areas
- Similar to one-tailed t-test
- Step 1:
  - If $K_{growth} > K_{gap}$:
    - Go to step 2
  - Else
    - P-value > 0.5 and we cannot reject the null
- Step 2:
  - P value is half as much as non directional alternative hypothesis
  - P value = 0.015/2 = 0.0075

Multiple testing

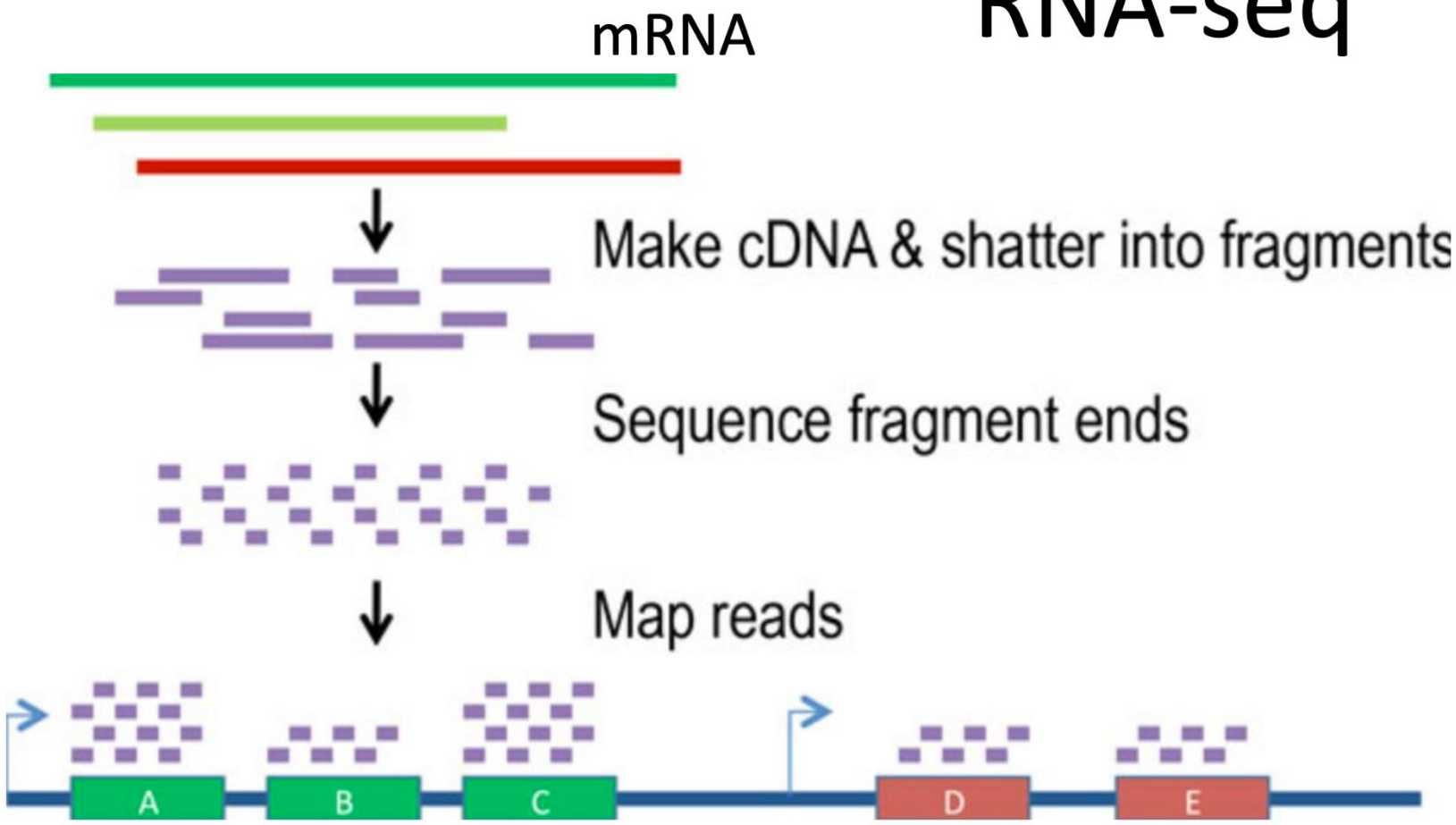
- Suppose we set $\alpha = 0.05$
  - aka we want our type I error to be around 95%
  - We want to reject the null hypothesis when it's true 5% of the time
- Say we do 100 independent tests and assume the null hypothesis is true for all 100 tests; how many times do we expect to reject the null hypothesis?
  - 100 tests * 0.05 = 5
    - (Poisson distribution with mean 5)
  - Since the null is true, these are all type I errors
  - P(reject at least one test) = $1-(1-0.05)^{100} = 0.994$
    - Nearly 100%! We're rejecting the null hypothesis (when true) much more than 5%
- A p value that is significant for one test is quite common for many tests

Similar examples

- Lottery
  - Chances for you to win are very low
  - But someone wins
  - Many, many people play
- Flipping a coin
  - 10 heads in a row
  - 1/1024
  - Only need ~ 1000 individuals to expect it to happen to one person

Example

- The amount of mRNA is proportional to the expression level of the gene it codes for
- Use RNA-seq to compare gene expression between healthy and cancer cells



Genomics

	healthy cell #1	healthy cell #2	healthy cell #3	healthy cell #4	cancer cell #1	cancer cell #2	cancer cell #3	cancer cell #4
gene #1	2,004	2,010	2,030	2,025	2,015	2,003	2,006	2,016
gene #2	1,500	1,550	1,607	1,525	1,530	1,590	1,556	1,575
gene #3	3,054	3,044	3,037	3,050	3,048	3,059	3,060	3,044
gene #4	1,004	1,007	1,009	1,010	3,112	3,100	3,099	3,087
gene #5	1,567	1,588	1,590	1,550	1,600	1,580	1,594	1,544
…								

Do a separate t-test for all 20,000 genes

RNA-seq example

- Suppose the null hypothesis is true for all 20,000 genes
  - There's no difference between healthy and cancer cells
- Despite this assumption, how many significant results do we expect? (reject at α = 0.05)
  - 20000*0.05 = 1000!
  - It's probably that some genes have different expressions between healthy and cancer cells
    - But it would be hard to distinguish from false positives!

Bonferroni correction

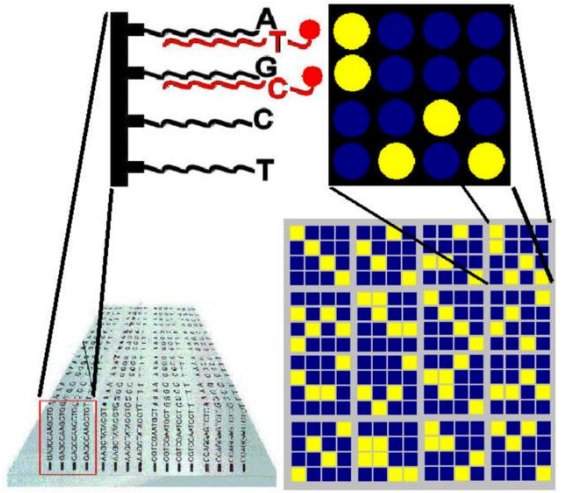
- Assume null hypothesis is true
- Let F_i be the probability that for test i we make a false positive error
- Then the probability we make at least one false positive error is:
  - For 2 tests:
    - $P(F_1 \; or \; F_2) = P(F_1) + P(F_2) - P(F_1 \; and \; F_2) \leq P(F_1) + P(F_2) = 2\alpha$
  - For n tests:
    - $P(F_1 \; or \; F_2 \; or \; \ldots or \; F_n) \leq P(F_1) + P(F_2) + \cdots + P(F_n) = n\alpha$

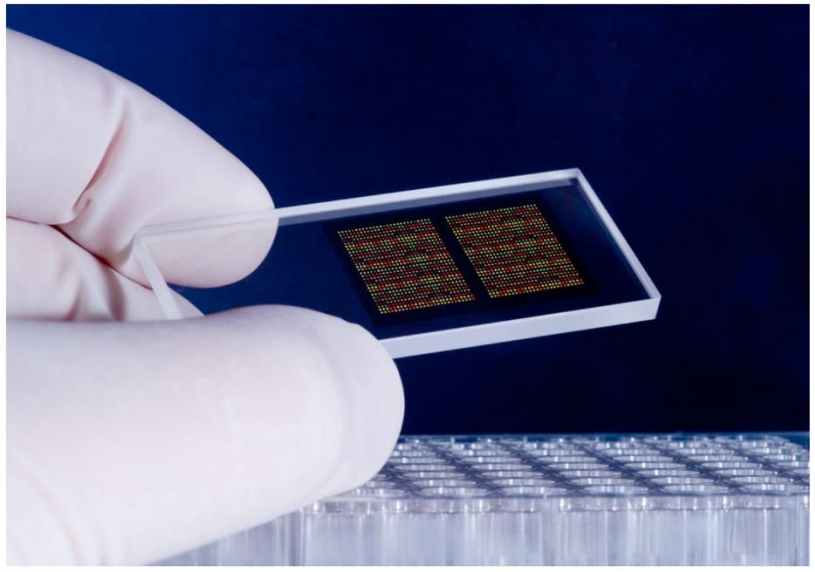
Bonferroni correction

- To keep the type I error below α, we **conservatively** require a p-value to be below (α/n) in order to reject the null hypothesis
- E.g. For α = 0.05
  - 4 tests, $\alpha/4$ = 0.05/4 = 0.0125
  - 100 tests, $\alpha/100$ = 0.05/100 = 0.0005

GWAS example

- Genome-wide association study (test)
- Testing variants (SNPs) across the genome against a phenotype
  - At a particular position on a chromosome, 90% of samples may have an A and 10% may have a T
  - Millions of SNPs across the entire human genome
- A single chip can identify millions of variants





The following SNP is strongly associated with the disease

	AA	AC	CC
Cases	5	295	700
Controls	990	10	0

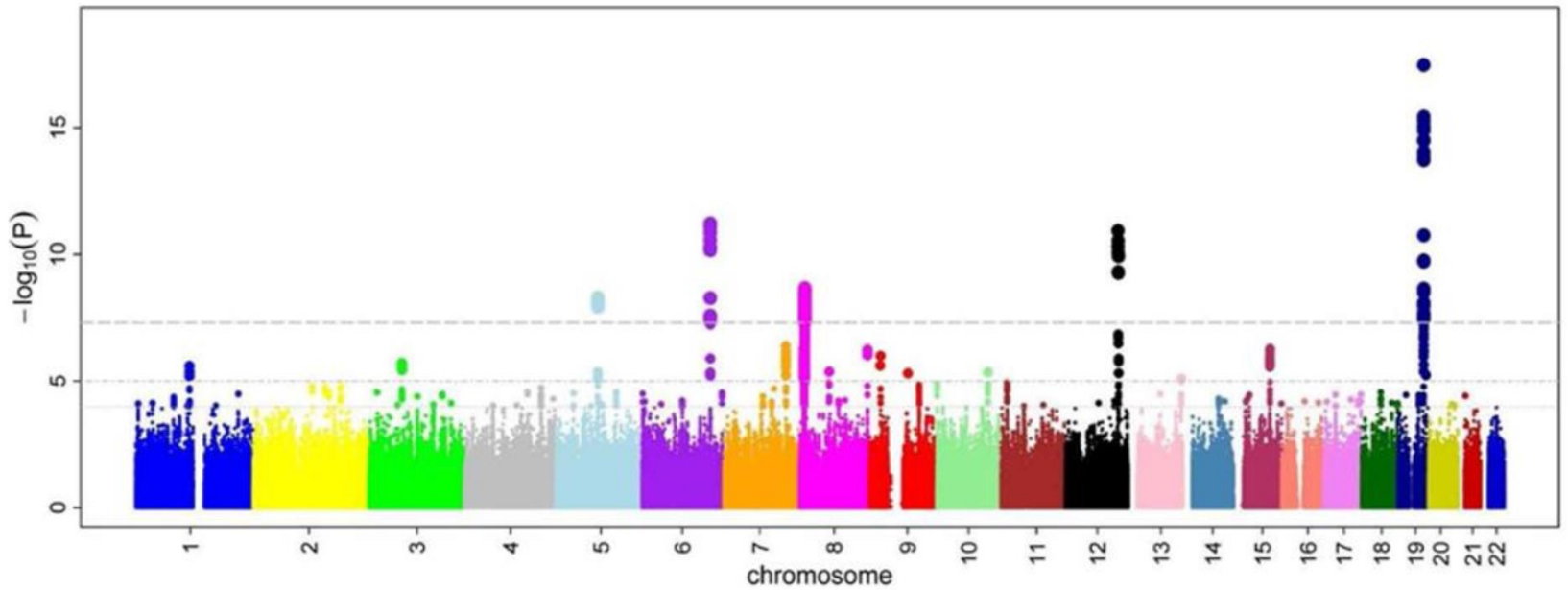
The following SNP is not associated with the disease

	CC	CT	TT
Cases	810	180	10
Controls	800	185	15

Bonferroni correction for GWAS

- Suppose we test each SNP for an association with a disease
  - 1 million tests
- Then $\alpha/1000000$ = 0.05/1000000 = $5x10^{-8}$

Manhattan plot



FDR – false discovery rate

- Genomic studies often perform thousands/millions of tests
- Bonferroni correction can be too conservative
- Instead of limiting the probability of making at least one type I error (like Bonferroni), another option is to limit the proportion of tests that are false positives

	Null true	Null false	Total
Don’t reject null	U (correct)	T (false negatives)	M-R
Reject null	V (false positives)	S (correct)	R
Total	M_0	$M-M_0$	M

- M total tests
  - Null is true for M_0 tests
  - Reject the null for R tests
  - V of the R tests are false positives
- The false discovery rate equals the expected value of V/R
  - The fraction of significant results (rejected) that are false positives
- Know that there is a way to restrict this

- One way to do science is:
  1. Make a hypothesis
  2. Collect some data to test this hypothesis
  3. Do a statistical test and report the p-value

- For this approach p-values work great

- e.g., Test the hypothesis that people who regularly skip breakfast weigh more than people who regularly eat breakfast (observational study so be aware of confounding variables)

- But you can imagine a more vague hypothesis that regularly skipping breakfast is bad for you and test all of the following things (each of which can be tested multiple ways):
  1. Increases weight
  2. Increases diabetes risk
  3. Increases heart disease risk
  4. Increases cancer risk
  5. Negative impact on mood and energy
  6. Decreases cognitive function
  7. Triggers hair loss
  8. Causes migraines

- One possible result is that all of these hypotheses are significant
- A very different result is that a research group kept testing different hypotheses until they found one that was significant

P-hacking

- As we've discussed, if you do enough statistical tests you will eventually find one with a low p-value

- Brian Wansink was a famous food science professor at Cornell University. Much of his research involved analyzing surveys. He and his research team would analyze these studies many, many different ways until they found something surprising ("Work hard, squeeze some blood out of this rock, and we will see you soon")

- He lost his job and had to retract numerous published papers over this (and other issues)

- In genomics (and other fields) recent technological advances allow us to test thousands or even millions of hypotheses at once

- e.g., Separately test all the genes to see if any are differentially expressed in people who regularly skip breakfast and people who regularly eat breakfast

- One solution then is to correct the p-values for the number of tests performed (Bonferroni or FDR)

- So if you do say 20 tests, you should report that you did 20 tests (and not pretend that the one test with the lowest p-value is the only one you did)

- To be significant, the p-value has to be lower than if you only did one test

- But what if 20 research groups (unbeknownst to each other) each do a similar experiment
- One group doesn't know about the other groups so they can't correct for multiple tests
- How many groups do we expect will get significant results (reject null at significance level $\alpha = 0.05$)?

Publication bias

- About 20 (0.05) = 1
- And this significant result from the one group is more likely to get published than the 19 non-significant results from the other groups

- Often when other research groups try to replicate published significant results they are unsuccessful
- It does not necessarily mean that any of the research groups did anything wrong
- It is important that different research groups try to replicate scientific findings

- Collect one data set and do many statistical tests on it. Then collect an independent data set. Repeat only those tests with low p-values on the second data set (to test whether those results are spurious or not)

- A similar approach is to just collect one big data set. But only search in part of the data (say one-half), and then repeat only those tests with low p-values on the other part of the data